

Sensitivity Analysis Audit Report

Project: Shadow AI Among Physicians in the German Healthcare System

1. Global Model Performance Evaluation Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only	N	Pseudo-R ²	AIC	BIC
0	0	0	0	320	0.1366	407.03	452.25
0	0	0	1	320	0.0922	377.37	422.59
0	0	1	0	320	0.1213	409.79	447.47
0	0	1	1	320	0.0891	374.56	412.25
0	1	0	0	320	0.1366	407.03	452.25
0	1	0	1	320	0.0922	377.37	422.59
0	1	1	0	320	0.1213	409.79	447.47
0	1	1	1	320	0.0891	374.56	412.25
1	0	0	0	320	0.1266	401.42	427.80
1	0	0	0	320	0.1266	401.42	427.80
1	0	0	1	320	0.0713	375.50	401.88
1	0	0	1	320	0.0713	375.50	401.88
1	0	1	0	320	0.1121	403.88	422.72
1	0	1	0	320	0.1121	403.88	422.72
1	0	1	1	320	0.0680	372.77	391.61
1	0	1	1	320	0.0680	372.77	391.61
1	1	0	0	320	0.1266	401.42	427.80
1	1	0	0	320	0.1266	401.42	427.80
1	1	0	1	320	0.0713	375.50	401.88
1	1	0	1	320	0.0713	375.50	401.88
1	1	1	0	320	0.1121	403.88	422.72
1	1	1	0	320	0.1121	403.88	422.72
1	1	1	1	320	0.0680	372.77	391.61
1	1	1	1	320	0.0680	372.77	391.61

2. Granular Model Interrogation Parameters

Model Specification 1

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.552	0.349	-1.579	0.114	0.576	0.291	1.142
KI-Bereitstellung	0.608	0.289	2.103	0.035	1.837	1.042	3.238
Guidelines	-0.948	0.507	-1.870	0.061	0.387	0.143	1.047
Schulung	0.380	0.362	1.051	0.293	1.462	0.720	2.971
Krankenhaus	0.526	0.297	1.770	0.077	1.691	0.945	3.026
MVZ	0.726	0.446	1.626	0.104	2.067	0.862	4.957
20-29 Jahre	2.201	0.814	2.702	0.007	9.033	1.831	44.576
30-39 Jahre	0.650	0.380	1.712	0.087	1.915	0.910	4.030
50-59 Jahre	-0.193	0.375	-0.515	0.606	0.824	0.396	1.718
60-69 Jahre	-1.085	0.440	-2.466	0.014	0.338	0.143	0.800
70 Jahre oder älter	-0.800	0.673	-1.188	0.235	0.449	0.120	1.681
männlich	0.077	0.260	0.297	0.767	1.080	0.649	1.798

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	60.575
LLR p-value	0.000
Pseudo R² (McFadden)	0.137
AIC	407.027
BIC	452.247
Hosmer-Lemeshow p	0.396
Accuracy	0.647

Model Specification 2

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.208	0.379	-3.186	0.001	0.299	0.142	0.628
KI-Bereitstellung	0.380	0.291	1.305	0.192	1.463	0.826	2.589
Guidelines	-0.275	0.526	-0.523	0.601	0.759	0.271	2.128
Schulung	0.431	0.364	1.184	0.236	1.539	0.754	3.143
Krankenhaus	0.023	0.328	0.071	0.943	1.024	0.539	1.945
MVZ	0.278	0.488	0.569	0.569	1.320	0.507	3.436
20-29 Jahre	1.458	0.573	2.543	0.011	4.296	1.397	13.214
30-39 Jahre	0.142	0.398	0.357	0.721	1.153	0.528	2.516
50-59 Jahre	-0.338	0.402	-0.839	0.401	0.714	0.324	1.569
60-69 Jahre	-1.784	0.570	-3.132	0.002	0.168	0.055	0.513
70 Jahre oder älter	-0.793	0.734	-1.080	0.280	0.453	0.107	1.907
männlich	0.589	0.275	2.144	0.032	1.802	1.052	3.088

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	35.871
LLR p-value	0.000
Pseudo R ² (McFadden)	0.092
AIC	377.373
BIC	422.592
Hosmer-Lemeshow p	0.135
Accuracy	0.709

Model Specification 3

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.386	0.339	-1.139	0.255	0.680	0.350	1.320
Krankenhaus	0.503	0.293	1.717	0.086	1.654	0.931	2.939
MVZ	0.591	0.440	1.344	0.179	1.806	0.763	4.275
20-29 Jahre	2.119	0.809	2.618	0.009	8.326	1.704	40.686
30-39 Jahre	0.548	0.373	1.468	0.142	1.730	0.832	3.596
50-59 Jahre	-0.275	0.369	-0.746	0.455	0.759	0.368	1.565
60-69 Jahre	-1.174	0.434	-2.704	0.007	0.309	0.132	0.724
70 Jahre oder älter	-0.932	0.666	-1.398	0.162	0.394	0.107	1.454
männlich	0.079	0.257	0.308	0.758	1.082	0.654	1.792
Governance_Index	0.186	0.148	1.258	0.208	1.204	0.902	1.608

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	53.813
LLR p-value	0.000
Pseudo R² (McFadden)	0.121
AIC	409.788
BIC	447.472
Hosmer-Lemeshow p	0.484
Accuracy	0.656

Model Specification 4

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.132	0.370	-3.059	0.002	0.322	0.156	0.666
Krankenhaus	0.020	0.325	0.062	0.950	1.021	0.539	1.931
MVZ	0.203	0.486	0.417	0.677	1.225	0.472	3.175
20-29 Jahre	1.413	0.570	2.481	0.013	4.110	1.346	12.551
30-39 Jahre	0.099	0.394	0.252	0.801	1.104	0.510	2.391
50-59 Jahre	-0.377	0.399	-0.944	0.345	0.686	0.314	1.500
60-69 Jahre	-1.842	0.568	-3.243	0.001	0.159	0.052	0.483
70 Jahre oder älter	-0.871	0.729	-1.194	0.233	0.419	0.100	1.749
männlich	0.585	0.274	2.131	0.033	1.794	1.048	3.071
Governance_Index	0.259	0.156	1.662	0.097	1.296	0.955	1.760

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	34.682
LLR p-value	0.000
Pseudo R² (McFadden)	0.089
AIC	374.562
BIC	412.245
Hosmer-Lemeshow p	0.215
Accuracy	0.712

Model Specification 5

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.552	0.349	-1.579	0.114	0.576	0.291	1.142
KI-Bereitstellung	0.608	0.289	2.103	0.035	1.837	1.042	3.238
Guidelines	-0.948	0.507	-1.870	0.061	0.387	0.143	1.047
Schulung	0.380	0.362	1.051	0.293	1.462	0.720	2.971
Krankenhaus	0.526	0.297	1.770	0.077	1.691	0.945	3.026
MVZ	0.726	0.446	1.626	0.104	2.067	0.862	4.957
20-29 Jahre	2.201	0.814	2.702	0.007	9.033	1.831	44.576
30-39 Jahre	0.650	0.380	1.712	0.087	1.915	0.910	4.030
50-59 Jahre	-0.193	0.375	-0.515	0.606	0.824	0.396	1.718
60-69 Jahre	-1.085	0.440	-2.466	0.014	0.338	0.143	0.800
70 Jahre oder älter	-0.800	0.673	-1.188	0.235	0.449	0.120	1.681
männlich	0.077	0.260	0.297	0.767	1.080	0.649	1.798

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	60.575
LLR p-value	0.000
Pseudo R² (McFadden)	0.137
AIC	407.027
BIC	452.247
Hosmer-Lemeshow p	0.396
Accuracy	0.647

Model Specification 6

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.208	0.379	-3.186	0.001	0.299	0.142	0.628
KI-Bereitstellung	0.380	0.291	1.305	0.192	1.463	0.826	2.589
Guidelines	-0.275	0.526	-0.523	0.601	0.759	0.271	2.128
Schulung	0.431	0.364	1.184	0.236	1.539	0.754	3.143
Krankenhaus	0.023	0.328	0.071	0.943	1.024	0.539	1.945
MVZ	0.278	0.488	0.569	0.569	1.320	0.507	3.436
20-29 Jahre	1.458	0.573	2.543	0.011	4.296	1.397	13.214
30-39 Jahre	0.142	0.398	0.357	0.721	1.153	0.528	2.516
50-59 Jahre	-0.338	0.402	-0.839	0.401	0.714	0.324	1.569
60-69 Jahre	-1.784	0.570	-3.132	0.002	0.168	0.055	0.513
70 Jahre oder älter	-0.793	0.734	-1.080	0.280	0.453	0.107	1.907
männlich	0.589	0.275	2.144	0.032	1.802	1.052	3.088

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	35.871
LLR p-value	0.000
Pseudo R² (McFadden)	0.092
AIC	377.373
BIC	422.592
Hosmer-Lemeshow p	0.135
Accuracy	0.709

Model Specification 7

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.386	0.339	-1.139	0.255	0.680	0.350	1.320
Krankenhaus	0.503	0.293	1.717	0.086	1.654	0.931	2.939
MVZ	0.591	0.440	1.344	0.179	1.806	0.763	4.275
20-29 Jahre	2.119	0.809	2.618	0.009	8.326	1.704	40.686
30-39 Jahre	0.548	0.373	1.468	0.142	1.730	0.832	3.596
50-59 Jahre	-0.275	0.369	-0.746	0.455	0.759	0.368	1.565
60-69 Jahre	-1.174	0.434	-2.704	0.007	0.309	0.132	0.724
70 Jahre oder älter	-0.932	0.666	-1.398	0.162	0.394	0.107	1.454
männlich	0.079	0.257	0.308	0.758	1.082	0.654	1.792
Governance_Index	0.186	0.148	1.258	0.208	1.204	0.902	1.608

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	53.813
LLR p-value	0.000
Pseudo R² (McFadden)	0.121
AIC	409.788
BIC	447.472
Hosmer-Lemeshow p	0.484
Accuracy	0.656

Model Specification 8

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.132	0.370	-3.059	0.002	0.322	0.156	0.666
Krankenhaus	0.020	0.325	0.062	0.950	1.021	0.539	1.931
MVZ	0.203	0.486	0.417	0.677	1.225	0.472	3.175
20-29 Jahre	1.413	0.570	2.481	0.013	4.110	1.346	12.551
30-39 Jahre	0.099	0.394	0.252	0.801	1.104	0.510	2.391
50-59 Jahre	-0.377	0.399	-0.944	0.345	0.686	0.314	1.500
60-69 Jahre	-1.842	0.568	-3.243	0.001	0.159	0.052	0.483
70 Jahre oder älter	-0.871	0.729	-1.194	0.233	0.419	0.100	1.749
männlich	0.585	0.274	2.131	0.033	1.794	1.048	3.071
Governance_Index	0.259	0.156	1.662	0.097	1.296	0.955	1.760

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	34.682
LLR p-value	0.000
Pseudo R² (McFadden)	0.089
AIC	374.562
BIC	412.245
Hosmer-Lemeshow p	0.215
Accuracy	0.712

Model Specification 9

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.694	0.346	2.004	0.045	2.002	1.015	3.947
KI-Bereitstellung	0.606	0.286	2.121	0.034	1.833	1.047	3.210
Guidelines	-0.887	0.496	-1.787	0.074	0.412	0.156	1.090
Schulung	0.323	0.358	0.901	0.367	1.381	0.685	2.785
männlich	0.102	0.256	0.398	0.691	1.107	0.670	1.830
Alter	-0.527	0.107	-4.917	0.000	0.590	0.478	0.728
Krankenhaus und MVZ	0.566	0.264	2.146	0.032	1.761	1.050	2.954

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.178
LLR p-value	0.000
Pseudo R² (McFadden)	0.127
AIC	401.424
BIC	427.802
Hosmer-Lemeshow p	0.198
Accuracy	0.656

Model Specification 10

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.694	0.346	2.004	0.045	2.002	1.015	3.947
KI-Bereitstellung	0.606	0.286	2.121	0.034	1.833	1.047	3.210
Guidelines	-0.887	0.496	-1.787	0.074	0.412	0.156	1.090
Schulung	0.323	0.358	0.901	0.367	1.381	0.685	2.785
männlich	0.102	0.256	0.398	0.691	1.107	0.670	1.830
Alter	-0.527	0.107	-4.917	0.000	0.590	0.478	0.728
Krankenhaus und MVZ	0.566	0.264	2.146	0.032	1.761	1.050	2.954

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.178
LLR p-value	0.000
Pseudo R ² (McFadden)	0.127
AIC	401.424
BIC	427.802
Hosmer-Lemeshow p	0.198
Accuracy	0.656

Model Specification 11

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.279	0.374	-0.747	0.455	0.756	0.364	1.573
KI-Bereitstellung	0.426	0.286	1.491	0.136	1.532	0.875	2.682
Guidelines	-0.250	0.509	-0.491	0.623	0.779	0.287	2.111
Schulung	0.386	0.356	1.083	0.279	1.471	0.732	2.958
männlich	0.562	0.269	2.087	0.037	1.754	1.035	2.975
Alter	-0.494	0.118	-4.197	0.000	0.610	0.485	0.769
Krankenhaus und MVZ	-0.005	0.291	-0.017	0.987	0.995	0.563	1.760

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	27.746
LLR p-value	0.000
Pseudo R ² (McFadden)	0.071
AIC	375.497
BIC	401.875
Hosmer-Lemeshow p	0.188
Accuracy	0.694

Model Specification 12

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.279	0.374	-0.747	0.455	0.756	0.364	1.573
KI-Bereitstellung	0.426	0.286	1.491	0.136	1.532	0.875	2.682
Guidelines	-0.250	0.509	-0.491	0.623	0.779	0.287	2.111
Schulung	0.386	0.356	1.083	0.279	1.471	0.732	2.958
männlich	0.562	0.269	2.087	0.037	1.754	1.035	2.975
Alter	-0.494	0.118	-4.197	0.000	0.610	0.485	0.769
Krankenhaus und MVZ	-0.005	0.291	-0.017	0.987	0.995	0.563	1.760

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	27.746
LLR p-value	0.000
Pseudo R ² (McFadden)	0.071
AIC	375.497
BIC	401.875
Hosmer-Lemeshow p	0.188
Accuracy	0.694

Model Specification 13

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.816	0.342	2.386	0.017	2.261	1.157	4.417
männlich	0.087	0.254	0.343	0.732	1.091	0.664	1.793
Alter	-0.536	0.107	-5.029	0.000	0.585	0.475	0.721
Krankenhaus und MVZ	0.511	0.260	1.968	0.049	1.668	1.002	2.776
Governance_Index	0.178	0.144	1.237	0.216	1.195	0.901	1.585

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.725
LLR p-value	0.000
Pseudo R² (McFadden)	0.112
AIC	403.877
BIC	422.719
Hosmer-Lemeshow p	0.342
Accuracy	0.656

Model Specification 14

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.816	0.342	2.386	0.017	2.261	1.157	4.417
männlich	0.087	0.254	0.343	0.732	1.091	0.664	1.793
Alter	-0.536	0.107	-5.029	0.000	0.585	0.475	0.721
Krankenhaus und MVZ	0.511	0.260	1.968	0.049	1.668	1.002	2.776
Governance_Index	0.178	0.144	1.237	0.216	1.195	0.901	1.585

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.725
LLR p-value	0.000
Pseudo R² (McFadden)	0.112
AIC	403.877
BIC	422.719
Hosmer-Lemeshow p	0.342
Accuracy	0.656

Model Specification 15

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.214	0.368	-0.583	0.560	0.807	0.392	1.660
männlich	0.550	0.269	2.047	0.041	1.733	1.024	2.933
Alter	-0.501	0.118	-4.266	0.000	0.606	0.481	0.763
Krankenhaus und MVZ	-0.027	0.289	-0.093	0.926	0.973	0.552	1.716
Governance_Index	0.267	0.151	1.770	0.077	1.306	0.972	1.755

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.474
LLR p-value	0.000
Pseudo R² (McFadden)	0.068
AIC	372.769
BIC	391.611
Hosmer-Lemeshow p	0.206
Accuracy	0.709

Model Specification 16

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.214	0.368	-0.583	0.560	0.807	0.392	1.660
männlich	0.550	0.269	2.047	0.041	1.733	1.024	2.933
Alter	-0.501	0.118	-4.266	0.000	0.606	0.481	0.763
Krankenhaus und MVZ	-0.027	0.289	-0.093	0.926	0.973	0.552	1.716
Governance_Index	0.267	0.151	1.770	0.077	1.306	0.972	1.755

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.474
LLR p-value	0.000
Pseudo R² (McFadden)	0.068
AIC	372.769
BIC	391.611
Hosmer-Lemeshow p	0.206
Accuracy	0.709

Model Specification 17

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.694	0.346	2.004	0.045	2.002	1.015	3.947
KI-Bereitstellung	0.606	0.286	2.121	0.034	1.833	1.047	3.210
Guidelines	-0.887	0.496	-1.787	0.074	0.412	0.156	1.090
Schulung	0.323	0.358	0.901	0.367	1.381	0.685	2.785
männlich	0.102	0.256	0.398	0.691	1.107	0.670	1.830
Alter	-0.527	0.107	-4.917	0.000	0.590	0.478	0.728
Krankenhaus und MVZ	0.566	0.264	2.146	0.032	1.761	1.050	2.954

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.178
LLR p-value	0.000
Pseudo R ² (McFadden)	0.127
AIC	401.424
BIC	427.802
Hosmer-Lemeshow p	0.198
Accuracy	0.656

Model Specification 18

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.694	0.346	2.004	0.045	2.002	1.015	3.947
KI-Bereitstellung	0.606	0.286	2.121	0.034	1.833	1.047	3.210
Guidelines	-0.887	0.496	-1.787	0.074	0.412	0.156	1.090
Schulung	0.323	0.358	0.901	0.367	1.381	0.685	2.785
männlich	0.102	0.256	0.398	0.691	1.107	0.670	1.830
Alter	-0.527	0.107	-4.917	0.000	0.590	0.478	0.728
Krankenhaus und MVZ	0.566	0.264	2.146	0.032	1.761	1.050	2.954

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.178
LLR p-value	0.000
Pseudo R ² (McFadden)	0.127
AIC	401.424
BIC	427.802
Hosmer-Lemeshow p	0.198
Accuracy	0.656

Model Specification 19

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.279	0.374	-0.747	0.455	0.756	0.364	1.573
KI-Bereitstellung	0.426	0.286	1.491	0.136	1.532	0.875	2.682
Guidelines	-0.250	0.509	-0.491	0.623	0.779	0.287	2.111
Schulung	0.386	0.356	1.083	0.279	1.471	0.732	2.958
männlich	0.562	0.269	2.087	0.037	1.754	1.035	2.975
Alter	-0.494	0.118	-4.197	0.000	0.610	0.485	0.769
Krankenhaus und MVZ	-0.005	0.291	-0.017	0.987	0.995	0.563	1.760

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	27.746
LLR p-value	0.000
Pseudo R ² (McFadden)	0.071
AIC	375.497
BIC	401.875
Hosmer-Lemeshow p	0.188
Accuracy	0.694

Model Specification 20

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.279	0.374	-0.747	0.455	0.756	0.364	1.573
KI-Bereitstellung	0.426	0.286	1.491	0.136	1.532	0.875	2.682
Guidelines	-0.250	0.509	-0.491	0.623	0.779	0.287	2.111
Schulung	0.386	0.356	1.083	0.279	1.471	0.732	2.958
männlich	0.562	0.269	2.087	0.037	1.754	1.035	2.975
Alter	-0.494	0.118	-4.197	0.000	0.610	0.485	0.769
Krankenhaus und MVZ	-0.005	0.291	-0.017	0.987	0.995	0.563	1.760

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	27.746
LLR p-value	0.000
Pseudo R ² (McFadden)	0.071
AIC	375.497
BIC	401.875
Hosmer-Lemeshow p	0.188
Accuracy	0.694

Model Specification 21

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.816	0.342	2.386	0.017	2.261	1.157	4.417
männlich	0.087	0.254	0.343	0.732	1.091	0.664	1.793
Alter	-0.536	0.107	-5.029	0.000	0.585	0.475	0.721
Krankenhaus und MVZ	0.511	0.260	1.968	0.049	1.668	1.002	2.776
Governance_Index	0.178	0.144	1.237	0.216	1.195	0.901	1.585

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.725
LLR p-value	0.000
Pseudo R ² (McFadden)	0.112
AIC	403.877
BIC	422.719
Hosmer-Lemeshow p	0.342
Accuracy	0.656

Model Specification 22

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.816	0.342	2.386	0.017	2.261	1.157	4.417
männlich	0.087	0.254	0.343	0.732	1.091	0.664	1.793
Alter	-0.536	0.107	-5.029	0.000	0.585	0.475	0.721
Krankenhaus und MVZ	0.511	0.260	1.968	0.049	1.668	1.002	2.776
Governance_Index	0.178	0.144	1.237	0.216	1.195	0.901	1.585

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.725
LLR p-value	0.000
Pseudo R² (McFadden)	0.112
AIC	403.877
BIC	422.719
Hosmer-Lemeshow p	0.342
Accuracy	0.656

Model Specification 23

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.214	0.368	-0.583	0.560	0.807	0.392	1.660
männlich	0.550	0.269	2.047	0.041	1.733	1.024	2.933
Alter	-0.501	0.118	-4.266	0.000	0.606	0.481	0.763
Krankenhaus und MVZ	-0.027	0.289	-0.093	0.926	0.973	0.552	1.716
Governance_Index	0.267	0.151	1.770	0.077	1.306	0.972	1.755

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.474
LLR p-value	0.000
Pseudo R² (McFadden)	0.068
AIC	372.769
BIC	391.611
Hosmer-Lemeshow p	0.206
Accuracy	0.709

Model Specification 24

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.214	0.368	-0.583	0.560	0.807	0.392	1.660
männlich	0.550	0.269	2.047	0.041	1.733	1.024	2.933
Alter	-0.501	0.118	-4.266	0.000	0.606	0.481	0.763
Krankenhaus und MVZ	-0.027	0.289	-0.093	0.926	0.973	0.552	1.716
Governance_Index	0.267	0.151	1.770	0.077	1.306	0.972	1.755

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.474
LLR p-value	0.000
Pseudo R ² (McFadden)	0.068
AIC	372.769
BIC	391.611
Hosmer-Lemeshow p	0.206
Accuracy	0.709